



WHITE PAPER: REQUIREMENTS FOR ENERGY STORAGE SYSTEMS USED IN DATA CENTERS

Revision 1.0, Version 1.0

January 2026



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OCP Ready™ Facilities Lead



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Revision History

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04 Dec 2024	1.0	0.1	Mark Dansie	draft
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Executive Summary

Over the past 13 years, a transformative shift has occurred in data center design, catalyzed by Facebook's groundbreaking decision to open-source its infrastructure blueprints and the subsequent formation of the Open Compute Project (OCP). The collaborative ecosystem of the OCP has fostered innovation across various aspects of data center architecture, notably introducing the Energy Storage Systems (ESS) into the Open Rack design. This decentralized form of uninterruptible power supply (UPS) took the shape of modular battery backup units seamlessly integrated within Open Racks. This approach offered granular power resilience at the point of need, enhancing availability and creating a more efficient use of power within the IT technical space to support critical IT loads.

The dominant technology underpinning these early ESS deployments has been lithium-ion based batteries. This chemistry presented compelling advantages over traditional ESS solutions, including higher energy density, longer cycle life, and improved efficiency. These benefits allowed for more compact and effective backup power within the tight constraints of an IT rack. However, the widespread adoption of lithium-ion based technology has also brought a critical consideration to the forefront: inherent fire risk. The thermal runaway characteristics of lithium-ion based batteries necessitate robust mitigation strategies at both the component and system levels.

This white paper will detail the worldwide codes and standards currently followed by data center operators deploying ESS and provide guidelines for safe design, operation, risk mitigation and maintenance. It will also explore the specific benefits and challenges associated with lithium-ion and other chemistries, and influence future revisions of the OCP Ready™ v2 for Hyperscale, OCP Ready™ for AI and other self certification requirements necessary to ensure the safe and reliable deployment of energy storage in data centers suitable for hyperscale wholesale and AI/ML infrastructure deployments.





Definitions

N/A

List of Abbreviations

ESS: Energy Storage System(s)

BESS: Battery Energy Storage System(s)

HESS: Hybrid Energy Storage System(s)

NiZn: Nickel-Zinc

VRLA: Valve Regulated Lead-Acid

TPPL: Thin-plate Pure Lead Batteries

Li-ion: Lithium-ion

LFP or LiFePO₄: Lithium Iron Phosphate

NMC or LiNiMnCoO₂: Lithium Nickel Manganese Cobalt Oxide

ANB: Aqueous Nickel Batteries

SIB and Na-ion: Sodium-ion Batteries

AHJ: Authority Having Jurisdiction

PPE: Personal Protective Equipment





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1. Introduction

This white paper explores the requirements for Energy Storage Systems (ESS) deployments within data centers, focusing on the critical considerations for safe and reliable deployment in hyperscale wholesale environments. As data center design evolves, integrating modular battery backup units directly into Open Racks, disaggregated power rack systems (e.g., side cars), and ESS-based UPS systems have become a key strategy for enhancing power resilience. A BESS is an electro-chemical system that provides a range of risks depending on the underlying battery chemistry that is employed. For example, while lithium-ion based batteries have been widely used in early BESS deployments due to their energy densities and efficiencies, their inherent fire risk necessitates stringent mitigation strategies. This document will therefore detail the pertinent codes and standards, explore the benefits and challenges of various battery chemistries, and outline the OCP Ready™ requirements crucial for ensuring the secure and effective integration of energy storage in modern data center infrastructure.

2. Compliance with Open Compute Project Tenets

Openness

This white paper openly details the requirements for BESS, including the benefits and challenges of various battery chemistries, the relevant codes and standards, and strategies for risk mitigation in service. By sharing this comprehensive information, it fosters collaborative innovation, encourages feedback, and promotes transparent infrastructure design within the data center community, enabling broader adoption and refinement of safe and reliable BESS deployments.

Efficiency

This white paper enhances efficiency by providing an in-depth analysis of various BESS technologies, highlighting their performance characteristics and potential for optimized power utilization within data centers. By detailing the specific benefits and challenges of different battery chemistries, and examining relevant codes and standards, the paper enables data center operators and designers to make informed decisions that lead to safer, more energy-efficient and cost-effective BESS deployments.





Impact

This white paper significantly contributes to the "Impact" tenet by addressing the critical challenge of safe and reliable BESS deployment in data centers, particularly concerning lithium-ion based battery fire risks. By detailing relevant codes, standards, and OCP Ready™ requirements, it directly influences industry best practices, leading to safer installations, reduced operational risks, and enhanced business continuity for hyperscale wholesale deployments.

Scale

This white paper supports the "Scale" tenet by providing standardized guidelines and requirements for ESS, particularly for hyperscale wholesale deployments. By detailing OCP Ready™ specifications, it enables large-scale, consistent, and interoperable ESS integration across numerous data centers, streamlining deployment processes and facilitating rapid expansion while maintaining safety and reliability across diverse operational environments.

Sustainability

This white paper directly supports the "Sustainability" tenet by advocating for the responsible and environmentally conscious deployment of ESS. By thoroughly analyzing various battery chemistries, including their ecological impacts and end-of-life considerations, and by detailing fire safety and operational best practices, the paper aims to minimize environmental risks and promote resource efficiency. Furthermore, by emphasizing compliance with relevant codes and standards, it encourages the adoption of sustainable technologies and practices, ultimately contributing to a more resilient, safe, and environmentally sound data center infrastructure.

3. Stakeholders, Use Cases, and Deployment Models

Stakeholders

Internal Stakeholders

- **Data-center designers and builders:** Responsible for integrating ESS into electrical architectures, mechanical layouts, and safety systems, ensuring compliance with IEC, UL, NFPA, and OCP standards.





- **Developers and operators:** Oversee day-to-day performance, monitoring, and maintenance of ESS assets to maximize uptime and efficiency.
- **Owners and executives:** Define investment priorities and total-cost-of-ownership (TCO) targets, balancing resilience with sustainability and capital efficiency.
- **Facility and operations engineers:** Interface with ESS equipment using appropriate PPE, battery management systems (BMS), fire-suppression systems, and OCP monitoring frameworks to maintain safety and availability.

External and Community Stakeholders

- **Tenants / prospective tenants:** Require assurances of power continuity and quality, safety certification, and service-level adherence, particularly in wholesale and hyperscale environments.
- **Investors and insurers:** Evaluate ESS adoption in the context of risk, asset valuation, and sustainability performance metrics (e.g., Scope 2 & 3 reporting).
- **Regulatory authorities and certification bodies:** Enforce and/or influence design and operational requirements through compliance mandates (UL 9540, IEC 62933, NFPA 855, etc.) and permitting.
- **Electric utilities, distribution system operators (DSO) and/or transmission system operators (TSO):** Study and approve ESS designs for grid interconnection, and safely and reliably operate the grid while customer ESS are interconnected.
- **First responders:** Respond to alarms related to ESS and other site hazards with proper training and personal protective equipment (PPE).
- **OCP community contributors:** Drive open-standard development, interoperability testing, and publication of reference designs that accelerate industry alignment.
- **Governmental authorities:** Government bodies which define legislation and new regulations and make public investment in R&D and infrastructure.(DOE)

Use Cases and Deployment models

- **Row- or Rack-Level (BBU and Sidecar)**
 - Use case: Provides shared energy buffering for several racks within a row or rack, supporting modular redundancy and balancing short-term load variations, there has been recent addition of super capacitors to smooth out power spikes by GPUs/TPUs.





- Deployment model: A common ESS is connected to a DC busway or Rack Power Distribution Unit (RPP), enabling distributed UPS functionality across multiple racks, but is not typically interconnected with the electric utility, DSO or TSO.
- **Facility-Level Energy (Centralized UPS, Containerized BESS, etc.)**
 - Use case: Provides primary back up power in the case of electrical loss from the grid and can support grid interaction through peak-shaving, demand-response, or renewable-energy smoothing when interconnected with the electric utility, DSO or TSO.
 - Deployment model: Large-scale ESS units (>100 kW)—typically located within mechanical rooms or containerized—interface with the main LV or MV switchgear.
- **Microgrid and Edge**
 - Use case: Enables sites to operate as hybrid microgrids, combining on-site generation (solar, fuel cell, or genset) with battery storage for grid support or islanded operation.
 - Deployment model: Centralized ESS with dynamic controls, synchronized with local generation assets and grid interfaces.

Charging Energy

In all ESS use cases and deployment models, facilities with ESS must plan for charging energy from a grid or local generation source. Unmanaged ESS charging load will be additive to IT and other power usage at the facility, which may exceed equipment and/or grid connection limits and may increase costs. Managing and integrating ESS charging energy requirements with the facility will minimize their impact on overall power usage effectiveness and service reliability.

4. Energy Storage Systems

There are several basic groups of ESS technologies in use today. They are usually broken into categories such as Mechanical, Gravity, Electrical, and Electrochemical.

In most Data Center applications, physical size for a given energy/power density is a major concern. Therefore, some energy storage technologies (such as Gravity) are rarely employed. Others, such as Mechanical (flywheels, etc.) are less common due to perceived issues with either





runtime or reliability. For the purposes of this paper, the focus will be on the most commonly used Electrochemical ESS technologies. Additional technologies may be added as they become commercially available and more widely applied in Data Center operations.

Design, construction and operational considerations for ESS are increasingly needed in the data center industry, with safety as the primary common principle. No battery or other ESS technology is inherently safe, and NFPA 70E (Standard for Electrical Safety in the Workplace) indicates an electric shock hazard is present at voltages greater than 50 volts. As ESS deployments in data centers move towards operating at 400-1500 volts with “hot swap” capabilities, there are significant risks that must be mitigated. Only trained personnel shall handle, service, and/or operate ESS using appropriate PPE and safety supervision. All ESS must be treated as energized equipment unless it has been properly de-energized and tagged for maintenance or disposal. Failure to follow such standards and procedures can result in significant property damage and data loss (Seoul, 2025), injury (Surprise, AZ 2019) or death.



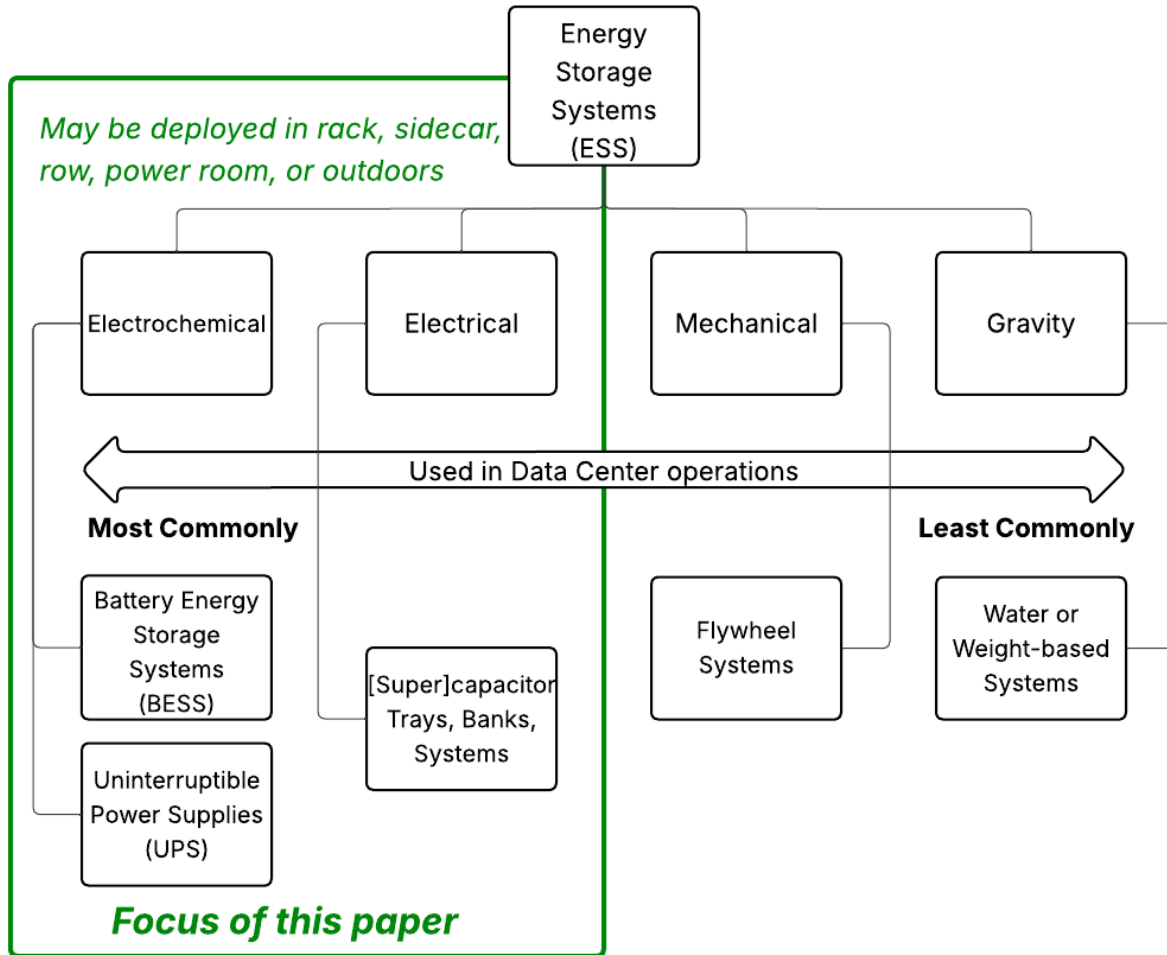


Figure 1. Focus area of this paper



ESS Architectures & Topologies

This section provides a technical classification of common ESS architectures and electrical topologies used in data-center environments. The intent is to describe how ESS may be structured from an electrical coupling, distribution, control, and grid-interaction perspective, independent of specific deployment use cases or battery chemistries. These architectural distinctions influence safety analysis, protection coordination, system control, and standards applicability, and are therefore relevant to compliance with internationally recognized codes, such as the ICC International Fire Code, NFPA 1 National Fire Code, NFPA 70 National Electrical Code, and standards such as NFPA 855, ANSI/CAN/UL 9540, and the IEC 62933 series.

NFPA 855 is the consensus bellwether document regarding the installation of ESS in both occupied and unoccupied locations. The 2026 revision provides the best guidance for installation and utilization safety for all ESS installations. This Standard provides the requirements for all aspects of ESS installation, including location, sizing, ventilation, fire suppression, explosion control, egress paths, storage, and both commissioning and decommissioning.

NOTE: NFPA 855 has, from its earliest publication, included a stipulation (Chapter 1, Paragraph 1.4.2) that if the AHJ determines that an existing situation presents an unacceptable degree of risk, he can retroactively apply any portion of 855 deemed appropriate.

NFPA 855 requires explosion control and prevention, where applicable, for electrochemical ESS, except for lead-acid and other aqueous technologies. Solutions may include NFPA 69 combustible concentration reduction systems or NFPA 68 venting, based on the hazard mitigation analysis and AHJ requirements.

The future NFPA 800, Battery Safety Code, will provide requirements for storage, transportation, recycling and disposal of batteries that may be used in ESS and other battery-powered data center equipment.





Table 1. ESS Architecture & relevant codes

Topology	Architecture	Description	Relevant Codes and Standards
Electrical Coupling	DC-Coupled ESS	ESS connected directly to DC distribution paths supplying IT or sub-distribution systems	NFPA 70, 855; IEC 62933-1; IEC 62933-5-1
	AC-Coupled ESS	ESS interfaced through power conversion systems at the AC level	NFPA 70, 855 ; IEC 62933-5-2
System Distribution	Distributed ESS	Energy storage segmented into multiple smaller systems across defined fault domains	NFPA 70, 855 ; IEC TS 62933-2-2
	Centralized ESS	Energy storage consolidated into a single or limited number of systems serving large load domains	NFPA 70, 855; IEC 62933-5-2
Control Architecture	Autonomous Control	Local ESS control with independent protection and limited system-level coordination	NFPA 70, 855; IEC TS 62933-2-2
	Coordinated / Hierarchical Control	ESS coordinated via site-level energy management or power control systems	NFPA 70, 855; IEC 62933-2-1; IEC 61850
Energy vs Power Orientation	Energy-Oriented ESS	ESS designed primarily for energy capacity and sustained runtime	NFPA 70, 855; IEC 62933-2-1
	Power-Oriented ESS	ESS designed primarily for high power output and short-duration response	NFPA 70, 855; IEC 62933-1
Hybrid Architectures	Hybrid ESS	ESS combining multiple storage technologies within a single integrated system	NFPA 70, 855; IEC TS 62933-2-2; IEC TR 62933-2-201
Grid Interaction	Backup-only ESS	ESS operated without active participation in grid services	NFPA 70, 855; IEC 62933-5-1
	Grid-Interactive ESS	ESS capable of interacting with utility grids or microgrids	NFPA 70, 855; IEC 62933-5-3; IEC 61850; Electric utility / DSO / TSO rules



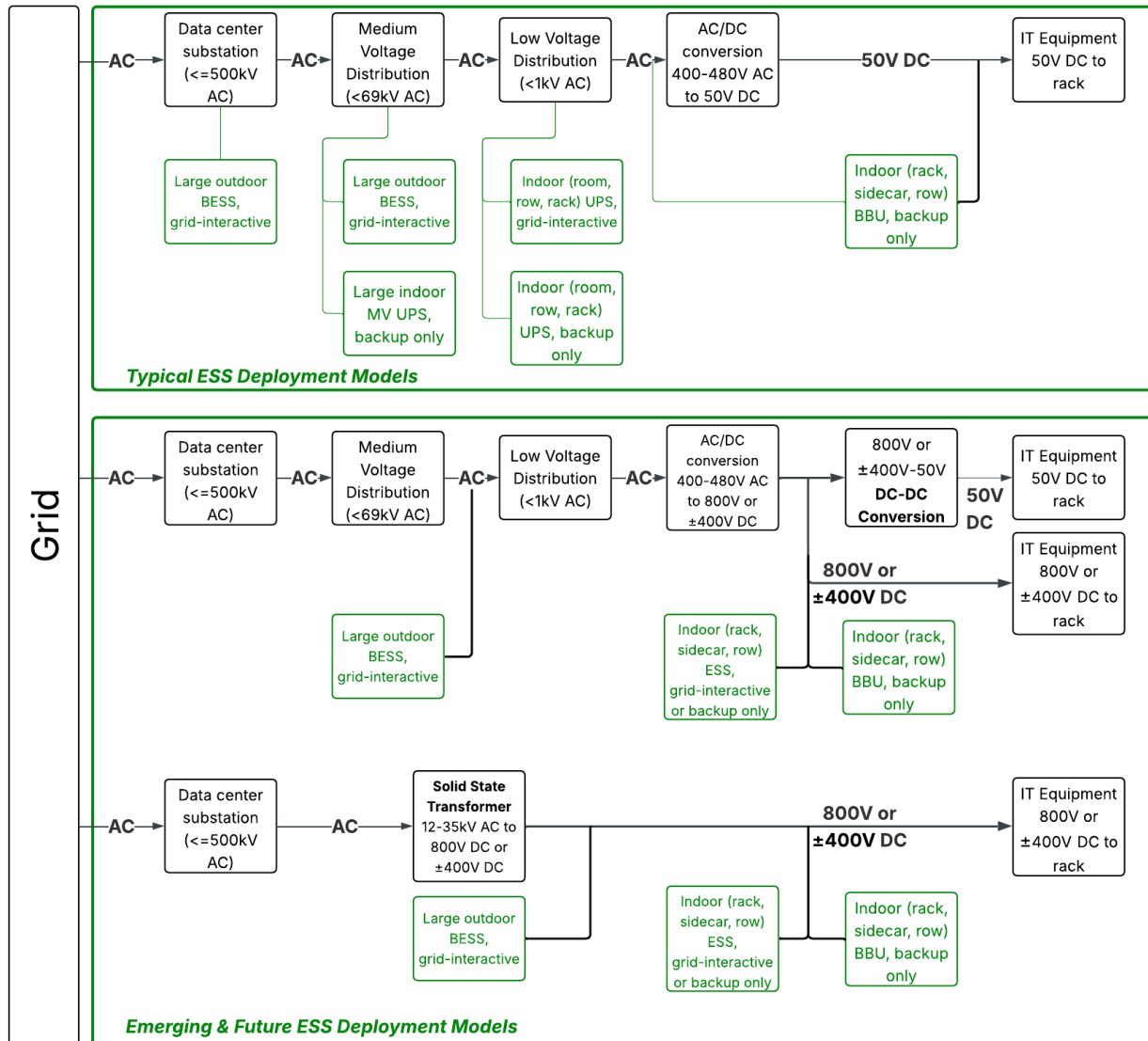


Figure 2. Illustrative Data Center Power Systems with ESS



ESS Technologies

Table 2. ESS Technologies

	Ceramic Capacitors (MLCC)	Electrolytic Capacitors (Al, incl. polymer/hybrid)	Film Capacitors (PP/PET DC-link)	Supercapacitors (EDLC / hybrid)	Li-ion –LTO, NTO, XNO	Na-Ion - NVPF	Na-Ion - Prussian White	Li-ion – NMC 811	Li-ion – LFP	Lead-Acid Batteries (VRLA/Flooded)
Primary role	Point-of-load HF decoupling, edge suppression	Bulk decoupling, hold-up energy	DC-link ripple handling, damping	Bridge power, pulse buffering, power quality	SD-ESS, UPS, BBU, fast charge/dischARGE buffering	High-power buffer / ride through	BESS	BESS, automotive	BESS, automotive	UPS ride-through + backup energy
Storage mechanism	Dielectric (ceramic)	Dielectric (electrostatic)	Dielectric (polymer film)	Electrochemical capacitor	Electrochemical	Electrochemical	Electrochemical	Electrochemical	Electrochemical	Electrochemical
Typical runtime	Nanoseconds → milliseconds	Nanoseconds → ~100 ms	Microseconds → tens of milliseconds	Sub seconds → second	Seconds → minutes	Seconds → minutes	Minutes → hours	Minutes → hours	Minutes → hours	Minutes → hours
Energy density	Very low	Low	Very low	Very low	Low–medium	Low	Medium	Very high	Medium	Low
Power density	Extremely high	High	Very high	Very high	Very high	Very high	Medium-high	High	Medium	Medium
Cycle life	Extremely high	High (if lightly stressed)	Very high	Extremely high (100k–1M+)	Very high (10k–20k+)	High-very high	Medium-high	Medium–high (thousands)	High (thousands to 5k+)	Low (hundreds)
Lifespan	Long (but can crack)	Medium	Long (often 15–25+ yrs)	Long (10–20+ yrs)	Long (10–20 yrs)	Long	Long	Medium–long (8–15 yrs)	Long (10–20 yrs)	Short (3–6 yrs)





Thermal / safety risk	Very low	Low	Very low	Low	Low-medium	Medium	Medium	Medium-high	Medium	Low
Maintenance	Very low	Low (but planned replacement)	Very low	Low (monitor balancing/health)	Low-medium	Low-medium	Low-medium	Low-medium	Low-medium	Medium
Response time band	sub-ns → ms	ns → 100 ms	tens of ns → ~10s of ms	μs → seconds	sub-ms → seconds	ms → seconds	ms → seconds	ms → seconds	ms → seconds	ms → seconds
Voltage rating practicality (ie for 800 V bus)	Low	Medium-low	Medium-high	Low-medium	High	High	High	High	High	High

Electrochemical Battery Energy Storage Systems

Valve regulated lead acid (VRLA) batteries

VRLA batteries (sometimes referred to as sealed lead acid batteries) depend on the oxygen recombination cycle for its operation and are a variant of the traditional flooded rechargeable lead acid batteries. Thin plate pure lead (TPPL) batteries are another variant of the VRLA design that may also be utilized in certain applications.

VRLA batteries are available either as a **gelled** battery or as an **absorbed glass mat (AGM)** battery. In a gelled VRLA battery the sulfuric acid electrolyte is mixed with fumed or colloidal silica to form a gel, whereas in the AGM VRLA battery the sulfuric acid electrolyte is kept absorbed in the microfiber glass mat.

The predominant failure modes of lead-acid batteries are internal corrosion of the grid material caused by exposure to the sulfuric acid electrolyte, and dry-out caused by the electrolyte loss during charging. Precise charge control is required to maximize life, but this is difficult to achieve in conventional series connected battery strings.





See NFPA 855 (2026) Annex B.5.2, IEC 60896-21/22 and IEC 62933-5-1 for hazard considerations.

Nickel zinc (NiZn) batteries

The NiZn battery is an aqueous alkaline chemistry that uses a potassium hydroxide solution as its electrolyte and combines the long life of nickel cadmium batteries and the high-energy associated with zinc electrodes. The positive electrode is typically nickel oxyhydroxide and the negative electrode is typically a zinc/zinc-oxide mixture.

A major advantage of the NiZn battery is that it is not susceptible to thermal runaway and, if constructed with a flame-retardant case and cover, it will not propagate an external flame. Instead, it will melt and self-extinguish.

Nickel-zinc batteries are manufactured in at least two formats; 1) cylindrical single cells, and 2) valve-regulated monobloc. Both formats utilize the same active materials and electrolyte, varying primarily in construction method.

The predominant failure mechanism of nickel-zinc batteries is active material depletion of the positive electrode. This is a much slower process than the erosion of the grid material in a lead-acid battery.

See NFPA 855 (2026) Annex B.5.5, IEC 62933-5-1 and IEC TS 62933-2-2 for hazard considerations.

Lithium-ion (Li-ion) batteries

Currently, several variants of Lithium batteries are commercially available, such as lithium iron phosphate (LiFePO₄ or LFP), lithium cobalt oxide (LCO), lithium manganese oxide (LMn₂O₄), lithium nickel manganese cobalt oxide (NMC), lithium nickel cobalt aluminum oxide (NCA) and lithium titanium oxide (LTO). Each variant is based on the type of positive electrode.

Lithium has several advantages that make it a key component in Li-ion rechargeable batteries. It is the lightest metal and most electropositive element; however, its high reactivity with water precludes the use of aqueous electrolytes in Li-ion batteries.





Each of the lithium technologies that are currently available has some level of potential to experience thermal runaway. The manufacturers of these products design the entire system to mitigate that potential and create a safe product for the intended application. It is critical that the entire system is installed and maintained in accordance with the manufacturer's instructions.

See NFPA 855 (2026) Annex B.5.3 and Annex G for hazard and fire suppression considerations, & IEC 62619, IEC 63056, and IEC 62933-5-2, supported by functional safety guidance in IEC TS 62933-2-2.

Sodium-ion based batteries

Several Na-ion chemistries are emerging, including sodium iron phosphate (NaFePO_4), sodium manganese oxide (NaMnO_2), sodium nickel manganese cobalt oxide (NaNMC), Prussian blue analogues (PBA), and hard-carbon (HC) anodes. Each chemistry varies mainly in the positive electrode composition and crystal structure, which define capacity, voltage, and cycle life.

See NFPA 855 (2026) Annex B.5.6 for hazard considerations.

Electrical

Supercapacitors

Supercapacitors are made using advanced carbon-based materials like activated carbon, carbon nanotubes, or graphene for their electrodes, leveraging high surface area for maximum charge storage. They function by storing energy physically—when voltage is applied, ions from the electrolyte align on the surface of porous electrodes, forming an electrostatic double layer, which allows rapid charging and discharging without chemical reactions, resulting in long cycle life.

Commonly, the electrodes are separated by a porous insulator and soaked in either aqueous or organic electrolytes, chosen based on cost, voltage requirements, and temperature stability. Variants are available from small modules (mini supercapacitors) to large industrial cells exceeding 5000 Farad, suitable for both backup and peak power applications in data centers.





This diverse construction allows supercapacitors to rapidly balance power surges, provide grid support, and extend the life of other backup systems.

Supercapacitors pose minimal fire hazard compared to lithium-based batteries since they lack flammable electrolytes and do not undergo thermal runaway, but poor quality units or electrical faults can still cause overheating or venting.

Hybrid supercapacitors that incorporate a lithium-ion-based cell with the capacitor portion will retain the inherent thermal runaway risk of the lithium-ion technology that is utilized.

See NFPA 855(2026) Annex B.6.1, IEC 62391, IEC 62933-5-1 & IEC TS 62933-2-2 for hazard considerations.

5. Insurance requirements /Best practices

Insurance companies like FM Global regularly updates its Property Loss Prevention Data Sheets with recommendations and best practices related to potentially hazardous equipment such as ESS, which is specifically addressed in its Data Sheets 5-28 and 5-33, with further guidance for Data Centers and related Facilities is in DS 5-32.

In addition, IEC 62485 series is also used as a main reference.

The following is the list of the key standards defining best practices for location selection, ventilation requirements and provision of alarm systems:

- FM Global Data Sheet 5-28 DC Battery Systems
- FM Global Data Sheet 5-32 Data Centers and Related Facilities
- FM Global Data Sheet 5-33 Lithium-ion Battery Energy Storage Systems
- IEC 62485-1 General Safety Information
- IEC 62485-2 Stationary Batteries





- IEC 62485-5 Safe Operation of Stationary Lithium-Ion Batteries
- IEC 60364-5-57:2022 Selection and erection of electrical equipment - Erection of stationary secondary batteries

Location

IEC 62485 standard installation location requirements for li-ion batteries are the same as TPPL and NiZn battery chemistries. FM Global Data Sheet 5-33 requires more onerous recommendations for Li-ion battery installation location. The options in the order of preferences are as follows:

1. In an enclosure outside and away from critical buildings or equipment in accordance with 2.3.2
2. In a dedicated building containing only LIB-ESS and associated support equipment in accordance with 2.3.3
3. In a dedicated exterior cutoff room that is accessible for manual firefighting operations and is constructed in accordance with 2.3.4
4. In a dedicated interior corner cutoff room with at least two exterior walls that is accessible for manual firefighting and is constructed in accordance with 2.3.4
5. In a dedicated interior cutoff room with at least one exterior wall that is accessible for manual firefighting and is constructed in accordance with 2.3.4



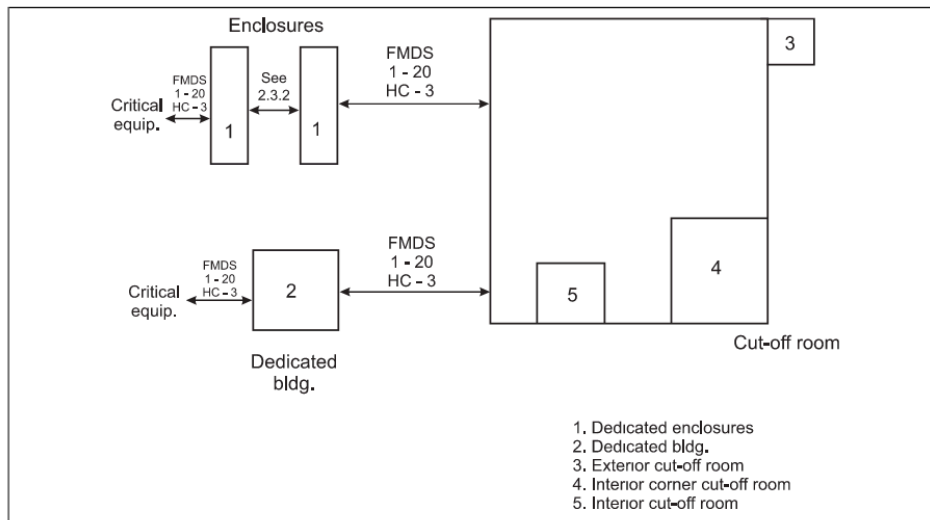


Figure 3. Best possible ESS location preferences

FM Global Data Sheet 5-33 section 2.3.4.3 recommends battery installation to be within a room with 2-hour fire rating as a minimum.

FM Global Data Sheet 5-33 section 2.3.5 recommends provision of separation distances for indoor battery rack installations:

- Provide 6 ft (1.8 m) minimum separation from the accessible face of an LIB-ESS rack to non-combustible materials, non-combustible construction elements, and the accessible faces of adjacent LIB-ESS racks.
- Provide 9 ft (2.7 m) minimum separation from the accessible faces of LIB-ESS racks to combustibles and combustible construction elements.
- Separation between non-accessible sides of adjacent racks should be determined by an installation-level fire test demonstrating that thermal runaway cannot propagate between racks. Where a test report is not available or the test did not result in a fire in the rack of origin, assume thermal runaway will propagate between racks.



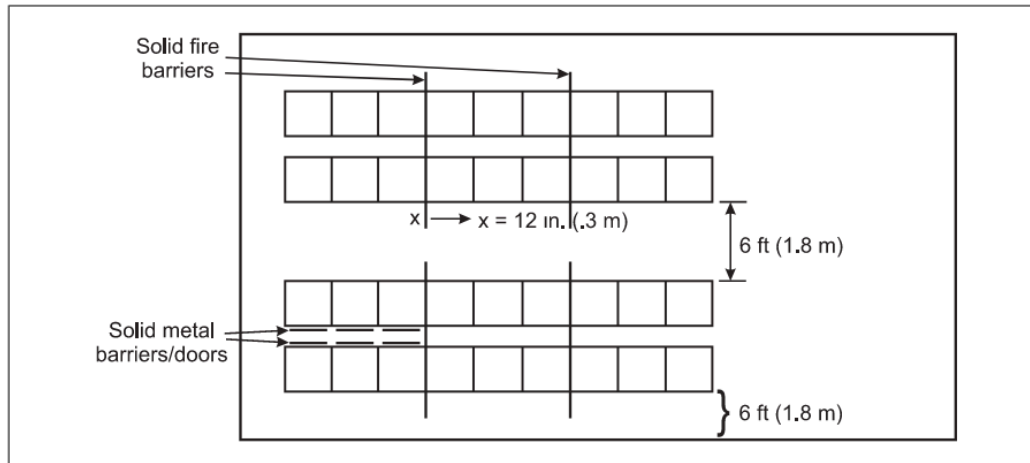


Figure 4. Thermal barriers to reduce fire risk area

Ventilation and Detection/Alarm

Hydrogen and oxygen gases can be released from batteries during charge, float charge and overcharge depending on chemistry. If high concentration is reached (4%vol), the mixture becomes explosive as implied in IEC 62485-2 section 7.2. It must be noted that gasses would continue to be emitted even after the charging is complete. The gas emission could continue for up to one hour or more depending on chemistry.

In case of a Lithium battery, the off-gassing is clearly an indication of a fault condition and preventative measures can be adopted to reduce the effect of catastrophic losses.

FM Global Data Sheet 5-28 section 2.3.1.3 declares gas limit of 1% of total room volume.

A hydrogen detection system or ventilation failure monitoring device is required to alarm remotely at a constantly attended location as per FM Global Data Sheet 5-28 section 2.3.1.4.





Continuous natural or forced mechanical ventilation is required to remove gases from the battery room. Ventilation is to be designed and provided in line with IEC 62485-2 section 7.2, 7.3 and Annex B. FM Global Data Sheet 5-28 section 2.3.1.3 declares recommendation to achieve 5.1l/s/m² of floor area (assumed floor area of the room) or provision of ventilation and thermal management in line with IEEE 1635.

FM Global Data Sheet 5-28 aligns with IEC 62485-2 in relation to ventilation and temperature monitoring provisions.

Online condition monitoring system is recommended for data center UPS systems and any type of batteries as detailed in FM Global Data Sheet 5-28 section 2.3.6.

FMG DS 5-33 specifically for LIB advises to incorporate an early intervention system to automatically and electrically isolate the LIB-ESS, using one of the following approaches:

1. High cell temperature: The cell manufacturer should provide the threshold temperature indicating an abuse condition based on 100% SOC. The majority of cells within a module should be constantly monitored.
2. Off-gas detection: Provide FM Approved gas detectors capable of detecting the volatile organic compounds associated with the off-gas event that precedes thermal runaway.

As per FM Global Data Sheet 5-33, Section 2.3.3.5, it is essential to maintain an air circulation rate of 0.3 m³/min/m² of floor area if the battery room is larger than 46.5m². This requirement could be met by the airflow provided through the cooling equipment, and would not necessarily require a dedicated supply and extract ventilation system.

In the event that the hydrogen levels exceed 10% of the Lower Explosive Limit (LEL) and/or thermal runaway, a hot-gas/hot-smoke emergency exhaust ventilation must be provided in accordance with Section 2.5.5.3. The required exhaust rate is 0.75 m³/min per m² of Floor area. For this calculation, if a dedicated battery room is not provided, the floor area could be defined as the physical footprint of the battery cabinets, plus an additional 1 metre in front and 0.5 metres behind each cabinet.

The emergency extract system should be configured to draw air from or near the ceiling, where hydrogen is likely to accumulate due to its low density. Exhaust air must be discharged to the





outdoors, directed away from air inlets, doorways, or other openings to prevent re-entrainment. Make-up air inlets should also be located on exterior walls, at positions remote from exhaust outlets to avoid drawing exhaust gases back into the building.

Protection

FM Global Data Sheet 5-28 section 2.4 recommends that for all battery chemistries other than lithium-ion, automatic sprinkler system is designed and provided in line with FM Global data sheet 3-26. The system is to be for HC-2 hazard classification and use quick response sprinklers.

Clean agent fire extinguishing systems can be provided as supplementary protection when there is a need to limit equipment and nonthermal damage. FM Global data sheet 4-9 to be referred to for further details.

FM Global Data Sheet 5-33, specifically for Lithium ESS, recommends instead an hazard classification HC-3 with a design rate of 0.30gpm/3500sqft for dry (pre-action) systems (12mm/min/330m2).

6. BESS Testing

Under NFPA 855, ESS must be tested and listed by a Nationally Recognized Testing Laboratory (NRTL), with test data, evaluation information and calculations provided to the AHJ in addition to other plans and information. That will include: fire and explosion test data (such as those produced in a UL 9540A test), Hazard Mitigation Analysis (HMA), calculations or modeling data to determine compliance with NFPA 68 (explosion protection by deflagration) and NFPA 69 (explosion prevention) as required, and other data or information if needed to support deviations from minimum safety requirements^[1].

[1] NFPA 855 (2026) Annex G1.3.2





Current Testing Methods

CSA/ANSI C800:25

CSA/ANSI C800:25 (consensus standard) and “CSA TS-800:24” (technical specification) is a testing protocol for the safety and quality assurance of ESS. It was created to ensure that these systems can withstand extreme conditions and perform safely over their operational lifecycle. In addition to UL 9540A (5th Ed.), the industry now has CSA/ANSI C800:25, which provides a consensus LSFT method. CSA TS-800:24 was a prior technical guidance document that was superseded by CSA/ANSI C800.

ANSI/CAN/UL9540A 5th Ed. (2025)

Test method designed to evaluate the risk of thermal runaway and fire propagation in battery energy storage systems. Essentially, it assesses how well a BESS can contain or mitigate the dangers of a battery fire. UL 9540A will expand the installation level test to encompass large-scale fire testing to assess ignited (forced) fire scenarios and propagation. The large-scale fire test method aligns with the NFPA 855 guidance on LSFT.

See UL Solutions (commercial arm of UL) <https://www.ul.com/services/ul-9540a-test-method>

NFPA 855 (2026)

NFPA 855 (2026) Annex G provides guidance on the scope and implementation of the Large Scale Fire Testing now required by NFPA 855 in 9.2.1. An LSFT is intended to fully evaluate the fire and explosion performance of an ESS, including by the ignition of vent gases, to determine the effects of a fire condition on surrounding exposed combustible materials.





7. Fire Suppression Systems

Fire suppression systems related to ESS must be designed by or in consultation with a Fire Protection Engineer, including an ESS-specific Hazard Mitigation Analysis or similar approach, as required by NFPA 855.

NFPA 855 (2026) 4.9.4.1*

Automatic fire control and suppression systems shall comply with the following standards, or their equivalent, unless modified in Chapters **9** through **17**:

- (1)NFPA 13 (Standard for the Installation of Sprinkler Systems)
- (2)NFPA 12 (Standard on Carbon Dioxide Extinguishing Systems)
- (3)NFPA 15 (Standard for Water Spray Fixed Systems for Fire Protection)
- (4)NFPA 750 (Standard on Water Mist Fire Protection Systems)
- (5)NFPA 770 (Standard on Hybrid (Water and Inert Gas) Fire-Extinguishing Systems)
- (6)NFPA 2001 (Standard on Clean Agent Fire Extinguishing Systems)
- (7)NFPA 2010 (Standard for Fixed Aerosol Fire-Extinguishing Systems)

For lithium-ion ESS, NFPA 855 (2026) Annex G - Guide for Suppression and Safety of Lithium-Ion Battery (LIB) Energy Storage Systems (ESS) “focuses on hazard identification and assessment, firefighting, fire protection, and fire and gas detection.”^[1]

Other ESS technologies and their fire, chemical, electrical, stranded/stored energy, and physical hazards are referenced in detail in Annex B – Battery Energy Storage System Hazards, subsection 5: Commercially Available Battery Technologies.^[2]

[1] NFPA 855 (2026) Annex G

[2] NFPA 855 (2026) Annex B.5





Automated Fire Control and Suppression

Water Based Systems

Water-based fire suppression will typically be installed in indoor areas containing ESS where personnel have regular access. These fire suppression systems are typically omitted in outdoor, containerized ESS that cannot be occupied.

Lithium-ion BESS with active, integrated water-based heat & fire suppression systems have shown that applying water to battery cells to prevent or reduce thermal runaway can result in electrical shorts and smoke, leading to subsequent water discharges and equipment damage (Vistra Moss Landing, 2021).

Lithium-ion (and other types when indicated by the HMA) battery ESS fires may be addressed by first responders by allowing thermal runaway to propagate through the ESS until the available fuel is consumed - a “let it burn” approach. Outdoor, containerized BESS fires are most likely to be treated this way.

Facilities containing ESS must integrate those systems and local first responders into overall fire and emergency planning, via a Hazard Mitigation Analysis (or similar study) and coordinated emergency response planning (section 11).

Gas Based Systems

Gas-based fire suppression systems use inert gases or other clean agents to suppress a detected fire event. For ESS, particularly lithium-ion BESS, water is typically the only recommended suppression agent, though it may not be applied by first responders, as discussed in the previous section. In addition, BESS failures can release gases that must be ventilated to the atmosphere before reaching critical levels that risk explosion.

Gas based fire suppression systems typically require closed ventilation systems to maximize effectiveness of the clean agent. Closed ventilation for clean agent suppression combined with BESS offgassing has previously contributed to an accident with serious injuries (McMicken BESS, Surprise AZ 2019). When clean agent systems are present for non-ESS fire suppression, these issues must be considered to ensure personnel safety.





8. Detection systems

Areas containing ESS will typically require smoke detection, thermal image fire detection, or radiant-energy-sensing systems in accordance with NFPA 72[1], designed for the specific ESS chemistry.

Detection Systems for lithium-ion based systems are addressed in detail in NFPA 855(2026) Annex G.7.3, and many of the approaches also apply to other ESS types. Those systems include:

- Battery Management System – can provide early warnings and critical data to first responders
- Smoke Detection – typical and aspirating (also known as Very Early Smoke Detection Apparatus or VESDA)
- Flame Detection
- Heat Detection
- Thermal Imaging
- Gas Detection – can pick up indications prior to and after thermal runaway events

9. Explosion Prevention/Control

ESS require passive and active protections for explosion control. Physical, electrical, fire, or other damage to electrochemical ESS can cause the release of large volumes of gases which may be toxic and/or explosive. Therefore, NFPA 855 requires all ESS mitigate these hazards.

Passive mitigations may include fixed ventilation and/or deflagration panels. Active measures include ventilation systems designed to exhaust gases to the atmosphere, which are considered a critical safety system for the ESS and must be provisioned with emergency backup power as noted below.





Equipment physically attached externally to an ESS, such as monitoring systems, HVAC units, or any other equipment should be considered an explosion hazard and mitigated to protect personnel and other infrastructure systems.

For large outdoor ESS, blast protection walls or barriers may be required by the AHJ to protect roads, buildings, or other areas.

All ESS shall be provided with a reliable explosion control and prevention system designed, installed, operated, maintained, and tested in accordance with NFPA 69 (Standard on Explosion Prevention Systems) (NFPA 855 (2026) 9.7.6.7.3).

10. Ventilation

In the case of a thermal event, offgas will be released that is both flammable and explosive. Managing and ducting this gas to avoid an explosion is a key challenge - particularly for indoor installations. Generally speaking a negative pressure ventilation system is preferred as it will result in reduced oxygen fed to the fire. There are many different physical setups and approaches that can be considered for mitigating risks - appropriateness depends on the technology, form factor, and the space within which it is installed. For instance, if a gas-based fire suppression medium is employed, then this must be taken into account with respect to the ventilation approach.

Some of the gases released will be lighter than air and some are heavier than air, and this complication is exacerbated by the fact that different events and system designs will produce gasses at a range of temperatures.

It is common in other industries, such as marine, to require explosion proof fans and equipment in the ventilation path due to the explosive nature of the gasses.

Backup power requirements for ventilation systems

All critical safety subsystems of an ESS, including ventilation systems, must be provisioned with emergency backup power (NFPA 855 4.1). For ventilation systems required to actively ventilate a space during an ESS fire event, backup power requirements may be 24-48 hours or more.





Temperature and humidity

ESS manufacturers specify temperature and humidity limits for equipment which must be followed to ensure safety, reliability, and longevity. Specifications may differ between ESS lifecycle stages; e.g. requirements for storage of non-operational equipment may vary from operational limits.

Some BESS use sealed, liquid-cooled battery modules. Coolant leaks leading to electrical shorts are a known source of BESS fires, requiring additional monitoring in those systems.

11. Handling of Lithium ion Batteries (LIB) Data Centers

Transportation

General Requirements

Dangerous goods - also referred to as Hazardous Materials - are articles and substances that pose hazards during transport. When transporting dangerous goods, the shipper must properly classify, package, mark, label, and document shipments in accordance with the applicable DG regulations. See DG regulatory standards below:

- Air Transport of DG: [IATA DG Regulations](#)
- Transport of DG To/From/Within the United States: [49 CFR Parts 100 - 180](#)
- Ocean Transport of DG: [IMDG Code](#)
- Highway Transport within Europe: [ADR Regulations](#)
- Transport within Canada: [TDGR](#)

Lithium-ion based Backup Battery Units (BBUs) incorporate lithium-ion batteries, typically with a Watt-hour rating above 300 Wh. As such, they are regulated in transport as a dangerous good and the shipper must follow specific packaging, marking, labeling, documentation, and training procedures.





All different lithium battery design types must successfully complete the UN Manual of Tests and Criteria, Part III, Sub-section 38.3 tests¹. The manufacturer of the cells and batteries must complete these tests and maintain records. Subsequent distributors of batteries must make available a UN 38.3 Test Summary document upon request.

Ground & Ocean Shipping of UN3480 Lithium Ion Batteries (Standalone, not in/with Equipment)

Packaging

For high-energy lithium-ion batteries (i.e., over 300 Wh), the shipper must securely package the batteries in an authorized UN Performance Package² unless the battery exceeds 12 kg and has an impact-resistant outer casing

Marks and Labels

Each package containing high-energy lithium-ion batteries must display:

- The Class 9 battery label
- UN3480 LITHIUM ION BATTERIES mark
- Address/Tracking Label

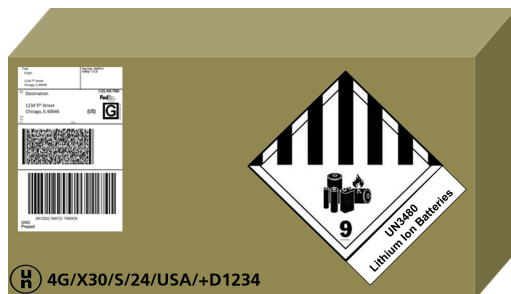


Figure 5: Lithium-ion packaging example

¹ <https://unece.org/about-manual-tests-and-criteria>

² 49 CFR 173.185(b)





Shipping Papers

Each shipment of high-energy lithium-ion batteries must be accompanied by a regulated shipping paper. For highway shipments, use of a Bill of Lading or alternative document is allowed. For ocean transport, the shipper must complete the IMO Declaration form.

Each shipping paper must contain:

- The basic description, which is the UN Number, Proper Shipping Name, Hazard Class, Packing Group.
 - For lithium-ion batteries, the proper description must read: UN3480, Lithium ion Batteries, Class 9
- Number and type of packages
- Total weight of hazmat being shipped
- Emergency response information & phone number
- A signed Shipper's Certification of compliance

The shipper must keep a copy of each shipping paper on file for at least two years

Training

Each employee involved in preparing and shipping lithium-ion batteries must complete applicable training³. In the United States, employees must complete recurrent training at least every three years. Be sure to maintain training records.

Storing

Best practices for on-site storage are found in the current International Fire Code, Section 320.

NFPA 855 defines standards for storage of lithium-ion batteries in Chapter 14 including requirements for collection areas, indoor and outdoor storage areas, fire prevention and mitigation, and explosion protection.

³ 49 CFR Part 172, Subpart H





Repairs and Replacement

Repairs to ESS must only be made by trained and qualified personnel with appropriate PPE, and must be documented in maintenance logs. If a repair uses other than identical or equivalent parts, it is considered a retrofit (NFPA 855 4.6.2). Any retrofits that use different ESS technology or chemistry are considered replacements and treated as new installations for compliance purposes (NFPA 855 4.6.4).

Recycling and End-of-life

NFPA 855 Chapter 8 requires owners of ESS to develop written decommissioning plans documenting how the BESS and all its supporting systems will be removed from the site.

When an ESS decommissioning plan is activated, the AHJ is notified, the plan is executed by trained and qualified personnel, and a final report documents its completion.

Ecological

ESS equipment contains environmentally hazardous materials and must not be sent to a landfill or resource recovery plant. Only dispose of ESS equipment via specialist licensed recycling and disposal contractors.

12. Hazard Mitigation Analysis and Emergency Response Planning

A Hazard Mitigation Analysis is now typically required for most ESS, with the exception of some aqueous chemistries, in NFPA 855 (2026) article 4.4.1 and evaluates the consequences of:

“(1) A thermal runaway or mechanical failure condition in a single ESS unit

(2) Failure of an energy storage management system or protection system that is not covered by the product listing failure modes and effects analysis (FMEA)

(3) Failure of a required system including, but not limited to, HVAC, liquid cooling system, BMS, communication system, or other critical systems that might impact normal operations





(4) Other failure modes required by the AHJ.”[1]

The HMA evaluates “the reliability and survivability of the following critical safety components or systems, during a thermal runaway propagation or single failure event:

(1) Exhaust ventilation

(2) Smoke detection

(3) Fire detection

(4) Fire suppression

(5) Combustible concentration reduction (CCR) system

(6) Gas detection

(7) Explosion control and prevention[2]

Emergency backup power must be provisioned to all critical safety systems, including the Battery Management System of an ESS.

Emergency Response Planning

NFPA 855 provides Annex G.10 “Guidance on Developing a First Responder Plan for LIB-Based ESS Installations” as a starting point for developing emergency response plans for a facility that includes ESS. The Standard requires working with local fire departments on preplanning and training for any incidents that may arise with ESS, particularly those containing lithium-ion batteries.

The Annex G.10 guidance on these Plans includes a framework and documentation requirements for a pre-incident emergency plan, including information needed by First Responders to an ESS incident. A comprehensive emergency plan is essential to ensure any ESS event is responded to safely by trained personnel with appropriate tools and PPE.





[1] NFPA 855 (2026) Article 4.4

[2] NFPA 855 (2026) 4.4.2.3

Data center personnel must only follow facility and ESS manufacturers' standard operating procedures and should never attempt to approach or extinguish an ESS in alarm that has visible indications of a fault - sparks, arcs, flames, gas plumes, etc. They must immediately evacuate the area, activate emergency plans, and wait for emergency response.

ESS fires present an immediate life-safety risk and no fire extinguisher exists that is effective on lithium-ion battery fires.

13. CSA Considerations

Civils

Civil engineering considerations for ESS must be made early in the facility design process. Large, outdoor containerized BESS require substantial land area adjacent to or nearby the supported facility and will typically be separated from buildings and other potential hazards. Such containers may weigh up to approximately 80,000 lbs (36,000 kg) and are installed on individual foundations into soils that must be geotechnically evaluated. Roads supporting transportation to and from the site and installation clearances and equipment should be considered. Some exposures may require blast protection with walls or other barriers.

Egress for onsite personnel who may be working in a BESS yard and access for first responders should be designed to local codes and in consultation with fire services.

Smaller ESS installed inside facility walls, whether in normally unoccupied or occupied spaces, will also need access for installation, repair, removal and emergency response. Other requirements may be indicated by the Hazard Mitigation Analysis or other engineering studies.





Structural

Structural design and engineering considerations for ESS are primarily driven by the weight of the systems. Indoor systems such as rack based lithium-ion BESS may weigh 4,000-6,000 lbs (1,800 - 2,700 kg) fully loaded. Such systems may need additional structural support in some facility designs. A Hazard Mitigation Analysis or other engineering study must also evaluate any structural exposure to a potential ESS fire - lithium-ion BESS fires can reach temperatures >2200F (1200C) which can weaken steel structures to failure (Moss Landing, 2025).

Outdoor systems deploying containerized BESS will treat each container as an unoccupiable structure, with foundations appropriate for the site - typically steel or concrete piles, or poured concrete pads. Anchoring is typically specified by the BESS manufacturer. In some locations, seismic protection of ESS will be required, with requirements typically specified in local codes and equipment tested & certified by the manufacturer.

Many structural requirements for ESS are defined in NFPA 855, Chapters 9.4 (Location Classification), 9.5 (Installation Size & Separation), 9.6 (Locations and Applications, including indoor & outdoor clearances and exposures) and 9.7 (Protection and Remediation).

Architectural

The civil, structural, and fire safety considerations discussed in previous sections all inform the architectural planning for a data center facility, to include the positioning of dedicated spaces for ESS, and considering ESS targeted for any shared (occupied) spaces. Integrated planning and design for ESS will benefit the long term safety and reliability of the facility and, in the event of an ESS failure and fire, mitigate property losses and ensure business continuity.

Segregated White Space/Grey Space

Isolation of Hazards

Separating BESS equipment from other systems and non-critical areas (whether located in white space with IT equipment or gray space with distribution equipment) should be considered to ensure operational integrity and hazard containment. Where possible, barriers should be designed





to support environmental and physical isolation in the respective white or gray space from other non-ESS equipment.

Cable Penetration Sealing Solutions

When implementing these segregated spaces, the choice of sealing solutions for cable and pipe penetrations becomes critical. Selecting systems that provide robust, multi-threat protection is paramount. Sealing solutions recognized by local certifying bodies are recommended. These may include nationally and regionally recognized authorities such as UL (Underwriters Laboratories), FM Approvals, and CSA (Canadian Standards Association). Sealing solutions should reinforce and maintain integrity of the structure or enclosure that they are being installed into.

Rodent Intrusion

Sealing solutions should be rodent proof. Some sealing solutions are tested and approved for animal mitigation, offering a robust defense against rodent damage and associated risks.

Environmental

Sealing solutions should protect against ingress from gas, water, and dust. This ensures the longevity and operational reliability of sensitive BESS equipment within both white and grey spaces. Consider smoke and gas sealing capability against toxic fumes in the case of fire. Watertight seals are also essential to contain liquid in case of fire suppression system activation.

Flexibility and Adaptability

The sealing system should be flexible enough to accommodate a wide range of cable and pipe sizes, simplifying design, installation, and future modifications. This adaptability is particularly beneficial in BESS environments where cable and pipe configurations can evolve.

Further details for CSA requirements in compliance with OCP requirements can be found [here](#).





14. Conclusion

ESSs used in data center operations are an essential part of the power management system that ensure reliable, stable, and efficient operations. Given these systems are typically electro-chemical products, data center owners and operators need to consider a wider variety of risks in the design process than other electronic equipment used in the data center to ensure safety and mitigate the potential for any adverse events. The information provided in this paper is intended to provide a foundation of knowledge for practitioners to consider as part of the design, permitting, and operational phases of a data center's lifecycle. With the advent of AI factories with ever increasing power densities that require more ESSs deployed throughout the data center, it becomes increasingly important to manage the electro-chemical risks these systems pose.





Appendixes

Appendix A. OCP Ready Requirements for ESS Matrix

Ref	Requirements Attribute (Optimum or Acceptable Parameter Required for OCP Ready Certification)	Acceptable Parameter	Optimum Parameter
2.3	Fire Suppression		
2.3-A	Automated Fire Control and Suppression	Must Comply with several NFPA Codes and Standards (or applicable equivalents)	Code requirements are the MINIMUM acceptable level of compliance.
2.3-B	Segregated White Space/Grey Space	Water based sprinkler system required for all electrochemical energy storage.	Design system with consideration for all flammable materials, including energy storage system. Some ESS may include flammable components that react negatively with certain suppression methods.
2.3-C	Gas/Smoke detection system	As required by applicable Fire Codes	Redundant systems that independently monitor and report/alarm on gas and/or smoke accumulation.
2.3-D	Ventilation	Required for all electrochemical energy storage systems.	Automated systems as usually preferred. Some ESS technologies require enhanced ventilation systems, and may require deflagration venting and/or explosion control are required in NFPA 68 and/or NFPA 69.
2.3-E	Energy Storage Technology	Should comply with capacity, location, spacing, and fire containment requirements in NFPA 855.	Select an energy storage technology based on lower fire risks and acceptance by FM Global or other liability insurance carrier.
2.3-F			





Appendix B. Codes and Standards

North America

Codes - In the US, safety Codes and Standards are primarily written by two organizations: The National Fire Protection Association (NFPA) and the International Code Council (ICC). The Codes and Standards are written by standing committees of experts in the subject field, and once published are adopted by governmental authorities (Federal, State and Local) as the mandated minimum requirements for new construction. The Codes and Standards may be amended by state or local governmental entities to elevate the requirements, but requirements may not be reduced.

Fire Safety:

Model Code:

These Codes are "model codes," meaning that they are designed to be adopted by local, state, or other jurisdictions. When a jurisdiction adopts one of these Codes, it becomes legally enforceable within that area.

Relationship with Other Standards:

These Codes often reference other industry standards, such as those published by UL, CSA, ASHRAE, ANSI, and many other Standards organizations. This means that to fully comply with the Code, you may also need to adhere to other relevant standards. A list of normative reference Codes, Standards and other documents that relate to the specific revision of the Code is published in each revision.

Energy Storage Systems:

The Codes have continually updated the rules regarding ESS, especially with the increased use of lithium-ion battery technology and other newer technologies. It should be noted that the Codes, as well as their normative references, are continuously updated as real-world circumstances change, and as such, it is imperative that one pay attention to any amendments or other supplemental





information that may be published in association with these Codes, Standards and other resources.

NFPA

National Fire Code (NFPA 1)

The National Fire Protection Association (NFPA) publishes the NFPA 1 National Fire Code on a three-year revision cycle. This Fire Code is accepted by eight US states. NFPA 1 is a "model code," meaning it is designed to be adopted by local, state, or other jurisdictions.

Although these Fire Codes are published regularly (2014, 2017, 2020, 2023, etc.), there are several jurisdictions that continue using earlier versions of the Code. When a jurisdiction adopts NFPA 1, it becomes legally enforceable within that area.

NFPA 1 references many "normative" standards from other organizations, which means that complying with NFPA 1 requires compliance with these other referenced standards.

NFPA has taken the various Codes and Standards regarding ESS and unified them into the NFPA 855 Standard for Installation of Stationary Energy Storage Systems.

National Electrical Code (NFPA 70)

NFPA 70 (NEC) is updated on a three-year cycle and is a Code with the purpose of "the practical safeguarding of persons and property from hazards arising from the use of electricity."^[1] It or another local electrical code must be followed along with NFPA 855 when designing and installing ESS. Many related Standards are correlated with the NEC, including NFPA 110, the Standard for Emergency and Standby Power Systems, NFPA 111, the Standard on Stored Electrical Energy Emergency and Standby Power Systems, and NFPA 70E, the Standard for Electrical Safety in the Workplace. Local jurisdictions adopt some or all of the NEC at their own pace, so versions older than the latest may be in effect in some places.





[1] NFPA 70, Article 90.2

Battery Safety Code (NFPA 800)

Future: a new Battery Safety Code (NFPA 800) is currently proposed in its first edition. Beyond NFPA 855, the proposed code will provide requirements for storage, transportation, recycling and disposal of batteries that may be used in ESS and other battery-powered data center equipment.

1.2.1 The purpose of this code is to prescribe minimum requirements necessary to establish a reasonable level of fire, life safety and property protection from the risks posed by electrochemical cell and battery fire and related electrical and deflagration hazards.

1.2.2 The code provides criteria for fire protection measures, emergency response consideration, and hazard control methods applicable to all lifecycle stages of batteries, except installation and operation of energy storage systems (ESS) covered in NFPA 855.

ICC

IFC

The International Fire Code (IFC) is published by the International Code Council (ICC) on a three-year revision cycle (2015, 2018, 2021, 2024, etc.). As with the NFPA 1 Code, there are several jurisdictions that are still using earlier revisions of the Code.

The IFC is accepted by forty-two US states, along with US Territories and protectorates. It is also accepted by several countries internationally.

As of the 2024 revision of the IFC, the International Code Council has started to align with the stipulations of NFPA 855, being careful to not infringe on the NFPA Copyright concerning its proprietary written material.





Relationship with Other Standards:

The IFC often references other industry standards, such as those published by the NFPA. This means that to fully comply with the IFC, one may also need to adhere to relevant NFPA standards. The IFC also references standards from many external organizations which will require compliance with those standards as well.

Energy Storage Systems:

The IFC has continually updated its rules regarding ESS, especially with the increase of lithium-ion battery technology.

Standards

Standards are primarily safety oriented and are developed by several different Standards Development Organizations (SDO). Among the most recognized and prolific of these are National Fire Protection Association (NFPA), UL Standards & Engagement (ULSE), part of the UL Enterprise (UL), Canadian Standards Association (CSA), & International Electrotechnical Commission (IEC), the American National Standards Institute (ANSI), and American Society of Heating and Refrigeration and Air Conditioning Engineers (ASHRAE). There are many other organizations that also contribute to the Standards base. One will find a list of those organizations and the relevant Standards published in each edition of each of the various Codes. Full Standards documents are typically available to read on each SDO's website, via their subscription services, and/or downloadable for a fee.

Underwriters Laboratories (UL) Standards & Engagement

UL Standards & Engagement is one of the most recognized SDOs and has published many Standards pertaining to fire, explosion, and battery safety. ULSE has published several of the Standards documents that both the NFPA and ICC Codes are based on. ANSI/CAN/UL 1973,





ANSI/CAN/UL 9540, and the ANSI/CAN/UL 9540A test method are directly aimed at the safety of Electrochemical Energy Storage technologies. These standards are harmonized with Canada and have been adopted by the US and Canada as the referenced safety standards. ANSI/CAN/UL 9540A has recently been revised and now includes a procedure to perform a Large-Scale Fire Test (LSFT) for ESS that are potentially capable of experiencing thermal runaway. Note: IEC 62933-5-2 highlights the importance of large-scale assessment of thermal runaway and references UL 9540A as an example of a test method used in practice.

CSA – Canadian Standards Association

CSA is another of the most recognized SDOs and has published many Standards that are applicable to ESS installations. While several of CSA's Standards are focused on the Canadian market, they have also published many that are directly or secondarily applicable to the US market. One of these is their recently published CSA C800:25 Large-Scale Fire Test (LSFT) procedure that provides an alternative LSFT methodology to the UL 9540A method.

ANSI – American National Standards Institute

ANSI is one of the primary certification organizations in the US. ANSI acts as co-publisher with ULSE and CSA for many of their Standards. Lending their authority to one of these Standards assures jurisdictions and authorities that the Standard is considered “best practice”.

International Standards

IEC

The International Electrotechnical Commission (IEC) is a global standards development organization that prepares international standards for electrical, electronic, and energy-related technologies. For Energy Storage Systems (ESS), the IEC focuses on system-level classification, performance, functional safety, electrical safety, environmental impact, and grid integration rather than prescriptive installation requirements. IEC standards are commonly adopted or transposed into regional and national frameworks (e.g., EN IEC, DIN EN IEC) and referenced by regulators





and certification bodies across EMEA. The **IEC 62933 series** represents the primary IEC standard family applicable to stationary ESS.

Overview of IEC 62933

IEC 62933 Parts	Title / Focus	Description
IEC 62933-1	General Requirements	Defines basic concepts, classifications, and terminology for energy storage systems (EES), providing a systematic approach to design and evaluation.
IEC 62933-2-1	Performance Testing for EES Systems	Specifies performance evaluation methods such as efficiency, response time, capacity, and lifecycle benchmarking.
IEC TS 62933-2-2	Functional Safety Assessment	A Technical Specification addressing system-level safety, identifying and mitigating hazards like thermal runaway or electrical faults.
IEC TS 62933-2-3	Reliability of Energy Storage Systems	Defines methods to assess reliability, covering failure rates, durability, and operational resilience over time.
IEC TR 62933-2-201	Guidance on Safety Cases	A Technical Report offering guidance for creating ESS safety documentation, connecting standards with real-world practice.





IEC 62933-4-2	Environmental Impact of EES Systems	Provides methods to evaluate environmental effects, considering carbon footprint, material efficiency, and recycling.
IEC 62933-4-4	End-of-Life Management	Establishes principles for decommissioning, recycling, and disposal, promoting circular economy practices.
IEC 62933-5-1	General Safety Considerations	Covers general electrical, chemical, mechanical, and fire safety requirements for stationary ESS across technologies.
IEC 62933-5-2	Safety for Large-Scale EES	Focuses on large battery ESS, covering thermal runaway prevention, emergency protocols, and system protection.
IEC 62933-5-3	Grid Integration Safety	Examines operational safety when connecting ESS to grids or microgrids, ensuring stability and safe integration.





Other Relevant standards which can be referenced

IEC Code	Title / Focus	Description
IEC 62619	Safety requirements for secondary lithium-ion batteries for industrial and UPS applications	Specifies safety testing and design for rechargeable lithium-ion cells and batteries used in industrial environments such as UPS, telecom, and energy storage. Emphasizes abuse testing, electrical, mechanical, and thermal protection for stationary battery systems.
IEC 62133	Safety requirements for portable rechargeable batteries	Focuses on safety for portable lithium-ion and nickel-based rechargeable batteries used in consumer electronics. Addresses chemical, mechanical, and electrical hazards, primarily for handheld or mobile devices.
IEC 60255	Measuring relays and protection equipment	Defines performance and functional requirements for protective relays used in electrical networks. Ensures reliable control, fault isolation, and system protection in both grid and data center environments.
IEC 61850	Communication networks and systems for power utility automation	Establishes standard communication protocols for automation of power systems, including Battery Energy Storage Systems. Defines mechanisms like GOOSE and MMS to ensure interoperability and reliable data exchange between devices.





IEC 61000	Electromagnetic compatibility (EMC)	Specifies EMC emission and immunity requirements for electrical equipment and ESS. Ensures equipment operates without causing or being affected by electromagnetic interference, harmonics, or noise disturbances.
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EMEA

Battery Passport

The EU Battery Regulation (2023/1542), adopted in July 2023, is designed to create a comprehensive framework covering the entire life cycle of batteries placed on the market in the European Union. The regulation covers all stages of the battery lifecycle including design, production, use, and end-of-life management. Its intent is to drive a sustainable and competitive battery supply chain for Europe while minimizing environmental and social impacts. The key provisions of the regulation include mandatory requirements for carbon footprint declarations (2026), due diligence for raw material sourcing (2027), recycled content targets for critical materials such as cobalt, lithium, and nickel (2031), and strict performance, safety, and labeling standards. The regulation also enforces collection and re-use/ recycling obligations to ensure circularity, particularly for critical materials. The regulation aims to make batteries not only safer and greener but also strategically aligned with the EU’s broader climate neutrality and resource resilience goals.

Regional

Deviation to standards: National and regional standards referenced in this section are based on IEC standards and are technically equivalent to their IEC counterparts. Any differences typically arise from national regulatory frameworks, installation codes, permitting processes, or enforcement practices rather than from modifications to the underlying IEC technical requirements.

NOTE: In addition to the internationally recognized standards of the IEC, many jurisdictions also refer to and follow many of the stipulations of both NFPA 855 and the ICC’s International Fire Code.





United Kingdom

In the United Kingdom, Energy Storage Systems (ESS) are designed in accordance with IEC standards adopted as BS EN IEC standards, with primary reference to the IEC 62933 series for system-level ESS classification and safety, and to BS EN IEC 62619 and BS EN IEC 63056 for lithium-based battery safety. These standards are applied alongside UK electrical installation requirements, including BS 7671 (IET Wiring Regulations), and are assessed by Building Control Bodies and other relevant Authorities Having Jurisdiction.

Fire safety for ESS in the UK is governed by a performance-based regulatory framework rather than a dedicated ESS-specific fire code. Key legislation includes the Regulatory Reform (Fire Safety) Order 2005, which mandates fire risk assessment and risk mitigation by the duty holder, and the Building Regulations 2010, particularly Approved Document B (Fire Safety). Supporting guidance such as BS 9999 (fire safety in the design and management of buildings) and BS 9991 (fire safety in residential buildings) may be applied where relevant. International standards and test methods, including NFPA 855 and UL 9540A, may be referenced as supporting evidence within a Fire Risk Assessment, but are not prescriptively mandated under UK law.

Germany

In Germany, Energy Storage Systems (ESS) are designed in accordance with IEC standards adopted as DIN EN IEC standards, with primary reference to the IEC 62933 series for system-level ESS safety and to IEC 62619 and IEC 63056 for lithium-based battery safety. Electrical installations are governed by national rules issued by DIN and VDE, including the DIN VDE 0100 series, which apply alongside the IEC framework.

Fire safety is regulated through building and fire-protection law, including the Musterbauordnung (MBO) and the corresponding Landesbauordnungen (LBO) at federal state level, supplemented by fire-protection concepts reviewed by local fire authorities. International standards such as NFPA documents or UL fire test reports have no legal standing and are not accepted compliance documents, but may be consulted as non-binding technical background information where national guidance for ESS hazards is still evolving.





Italy

In Italy, ESS design follows IEC standards adopted as CEI EN IEC standards, including the IEC 62933 series for system-level safety and IEC 62619 / IEC 63056 for lithium battery safety, applied together with national electrical rules issued by CEI.

Fire safety is regulated under D.P.R. 151/2011, which defines activities subject to fire-prevention controls, and the Codice di Prevenzione Incendi, issued through D.M. 3 agosto 2015 and maintained by the national fire authority (Vigili del Fuoco). These instruments establish the technical fire-prevention framework, approval process, and documentation requirements for ESS installations. International standards do not have legal status and are not required for compliance.

France

In France, ESS are designed in accordance with IEC standards adopted as NF EN IEC standards, including the IEC 62933 series, IEC 62619, and IEC 63056, applied alongside national electrical and safety regulations.

Fire safety is governed by the Code de la construction et de l'habitation (CCH) and national fire regulations. For public-access buildings, the Arrêté du 25 juin 1980 (ERP regulation) applies, while high-rise buildings are subject to the Arrêté du 30 décembre 2011 (IGH regulation). These frameworks define fire compartmentation, detection, suppression, and approval requirements for technical installations such as ESS. International standards may be referenced as technical background but have no regulatory standing.

Spain

In Spain, ESS installations reference IEC standards adopted as UNE-EN IEC standards, including IEC 62933 for ESS systems and IEC 62619 / IEC 63056 for lithium battery safety. Electrical installations are governed by the Reglamento Electrotécnico para Baja Tensión (REBT), approved by Real Decreto 842/2002, together with its associated technical instructions (ITCs).

Fire safety requirements are defined by the Código Técnico de la Edificación (CTE), specifically DB-SI (Seguridad en caso de Incendio), established by Real Decreto 314/2006. Fire-safety compliance for ESS is demonstrated through conformity with DB-SI requirements and authority





review. International standards are not formally adopted and may only be consulted as non-binding reference material.

Netherlands

In the Netherlands, ESS installations reference IEC standards adopted as NEN EN IEC standards, including the IEC 62933 series for system-level safety and IEC 62619 / IEC 63056 for battery safety. Electrical installations are governed by NEN 1010, which defines requirements for low-voltage electrical systems.

Building and fire safety are regulated under the Besluit bouwwerken leefomgeving (Bbl), which sets performance-based requirements for fire resistance, compartmentation, detection, and evacuation. ESS installations are assessed through municipal permitting processes. International standards such as NFPA documents or UL fire testing data are not formally adopted, but may be used as background technical information where relevant to fire-safety engineering justification.

Ireland

In Ireland, ESS design follows IEC standards adopted as I.S. EN IEC standards, including the IEC 62933 series for ESS systems and IEC 62619 / IEC 63056 for lithium battery safety. Electrical installations are governed by I.S. 10101:2020 – National Rules for Electrical Installations, published by the National Standards Authority of Ireland (NSAI).

Fire safety requirements are established under the Building Regulations 1997–2023, specifically Part B (Fire Safety), with compliance guidance provided by Technical Guidance Document B (TGD B). Fire-safety compliance for ESS is demonstrated through documented fire-safety design and assessment. International standards such as NFPA or UL documents have no regulatory status, but may be consulted as non-binding reference material in technical fire-safety studies.

Middle East (UAE, Saudi Arabia, Qatar & others)

In the Middle East, including the United Arab Emirates, Saudi Arabia, and Qatar, ESS designs commonly reference IEC standards directly, with application of the IEC 62933 series for system-level ESS safety and IEC 62619 / IEC 63056 for lithium-based battery safety. These standards are typically applied without formal national transposition.





Fire safety and permitting are governed by national civil defense and building-code frameworks, including the UAE Fire and Life Safety Code of Practice, the Saudi Building Code – Fire Protection (SBC 801), and Qatar Civil Defense regulations. These frameworks define fire protection, detection, suppression, and approval requirements for ESS installations. International standards such as NFPA documents may be referenced by authorities or consultants as technical background, but regulatory compliance is determined by local codes and approval processes.

APAC

The Asia–Pacific (APAC) region encompasses a wide range of regulatory frameworks; however, IEC standards form the primary technical foundation for Energy Storage Systems (ESS) across many APAC countries. Standards such as IEC 62933 (electrical energy storage systems), IEC 62619 (secondary lithium cells and batteries for industrial applications), and IEC 63056 (secondary lithium cells and batteries for use in ESS) are commonly adopted directly or referenced through national standards bodies. While the technical requirements are largely aligned with IEC, implementation, permitting, and enforcement vary by country through local electrical, fire-safety, and building regulations.

China

In China, ESS are governed primarily by national standards (Guo Biao – GB and GB/T) and mandatory regulatory requirements, with compliance determined exclusively against applicable Chinese standards. At the battery level, GB 31241-2022 defines mandatory safety and abuse testing requirements for lithium-ion cells and batteries. For ESS applications, standards such as GB/T 36276 (*Lithium-ion batteries for electrical energy storage*) and GB/T 34131 (*Technical specification for battery management systems for electrochemical energy storage*) establish technical, safety, and monitoring requirements at system level.

Chinese ESS standards are not direct adoptions of IEC standards, but many GB and GB/T documents are technically influenced by IEC principles and test methodologies, resulting in partial alignment in terminology and safety concepts while differing in structure, scope, and compliance mechanisms. Compared to IEC frameworks, Chinese standards are generally more prescriptive and closely tied to regulatory enforcement. US standards, including NFPA and UL documents,





have no regulatory status in China and cannot be used to demonstrate compliance, even though some test philosophies may appear functionally similar.

Japan

In Japan, Energy Storage Systems (ESS) are regulated primarily through national legislation and Japanese Industrial Standards (JIS). At the battery level, safety requirements for secondary lithium-ion batteries used in industrial and stationary applications are addressed in the JIS C 8715 series, which reflects IEC-based safety concepts adapted to Japanese regulatory requirements. Electrical equipment and installations associated with ESS are regulated under the Electrical Appliance and Material Safety Act (DENAN) and its associated technical ordinances, with conformity assessment obligations applying to designated products.

Fire safety and building compliance for ESS are governed by the Fire Service Act and the Building Standards Act. Under the Fire Service Act, ESS installations may be subject to provisions related to hazardous materials and fire-prevention equipment, with requirements enforced by local fire authorities in accordance with the Ordinance on Fire Prevention Equipment and Standards. The Building Standards Act establishes requirements for fire compartmentalization, fire resistance, evacuation, and emergency access, with detailed technical criteria defined in the Enforcement Order of the Building Standards Act. International standards, including IEC or NFPA documents, are not adopted as regulatory instruments; compliance is determined solely against Japanese law and JIS-based requirements, although international test data may be reviewed as background technical information during authority consultations.

Singapore

In Singapore, Energy Storage Systems (ESS) are designed and assessed using **international standards referenced through national regulations**, with **IEC standards** such as **IEC 62933** (electrical energy storage systems), **IEC 62619** (safety requirements for secondary lithium cells and batteries for industrial applications), and **IEC 63056** (secondary lithium cells and batteries for use in ESS) commonly used as technical references. Electrical installations are governed by **Singapore Standard SS 638** (*Code of Practice for Electrical Installations*), which defines requirements for low-voltage electrical systems and equipment.





Fire safety and building compliance for ESS are regulated under the **Fire Safety Act (Cap. 109)** and enforced by the **Singapore Civil Defence Force (SCDF)**. Fire-safety requirements are detailed in the **Fire Code 2018** (*Code of Practice for Fire Precautions in Buildings*), which specifies provisions for fire compartmentation, detection, suppression, access, and emergency response. ESS installations are subject to fire safety plan approval and inspection by SCDF. International standards, including NFPA and UL documents, are not adopted as regulatory instruments; compliance is determined against Singapore legislation and codes, although international test data may be reviewed as non-binding technical information during the approval process.

South Korea

In South Korea, Energy Storage Systems (ESS) are regulated through national legislation and Korean Standards (KS), with partial technical alignment to IEC concepts but independent regulatory enforcement. Safety requirements for lithium-ion cells and batteries used in stationary and industrial applications are addressed under KS C IEC 62619, which reflects IEC safety principles adapted to Korean conformity and certification processes. Electrical installations associated with ESS are regulated under the Electrical Appliances and Consumer Products Safety Control Act, including mandatory conformity assessment for designated equipment.

Fire safety and facility approval for ESS are governed by the Framework Act on Fire Services and its subordinate enforcement decrees and rules, administered by the National Fire Agency and local fire authorities. Building-related fire safety provisions are further regulated under the Building Act and associated enforcement regulations, covering fire compartmentalization, evacuation, detection, and suppression requirements. International standards, including IEC, NFPA, or UL documents, are not adopted as regulatory instruments; compliance is determined solely against Korean law and KS standards, with international material used only as non-binding technical background where appropriate.

Australia

In Australia, Energy Storage Systems (ESS) are implemented under a framework of Australian Standards and national building requirements. Electrical installations are governed by AS/NZS 3000 (Electrical Installations – Wiring Rules), which defines requirements for the design, construction, verification, and safety of electrical installations associated with ESS.





Fire safety and building compliance are regulated through the National Construction Code (NCC), with current consolidated editions such as NCC 2022 commonly referenced in practice. Fire-safety provisions within the NCC establish requirements for fire resistance, detection, suppression, and egress applicable to facilities containing ESS. Regulatory compliance is assessed by state and territory authorities under the NCC framework and applicable Australian Standards. International standards, including IEC, NFPA, or UL documents, are not mandated and may only be consulted as non-binding technical background information.

India

In India, Energy Storage Systems (ESS) are governed by national regulations and Indian Standards (IS) issued by the Bureau of Indian Standards (BIS), with technical alignment to IEC concepts depending on the subject area. Electrical installations and safety requirements are regulated under the Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, which establish rules for electrical safety, equipment, and installation practices.

Fire safety and building compliance for ESS are addressed under the National Building Code of India (NBC), with NBC 2016 widely referenced, including Part 4: Fire and Life Safety, which defines requirements for fire prevention, detection, suppression, compartmentalization, and evacuation. Enforcement is carried out by state and local authorities through permitting and inspection processes. International standards, including IEC, NFPA, or UL documents, do not have regulatory status for compliance and may be consulted only as non-binding technical background information.





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